

Design & RTL Implementation of a 4-Stage Pipelined Datapath

TECHNICAL PROJECT REPORT & ARCHITECTURAL DOCUMENTATION

Target Platform: Verilog RTL (Synthesis & Simulation)

Clocking Scheme: Two-Phase Non-Overlapping

1. Executive Summary

This report documents the architectural design, stage breakdown, and hardware verification of a high-performance 4-stage pipelined datapath implemented in structural Verilog. The core microarchitecture features an embedded 16-bit Arithmetic Logic Unit (ALU), a 16×16 -bit Multi-Port Register File, and a 256×16 -bit static SRAM-like synchronous Data Memory array. To safely maximize instruction throughput and eliminate race conditions across overlapping pipeline stages, the hardware leverages a robust, hardware-enforced **two-phase non-overlapping clocking scheme** (*clk1* and *clk2*).

2. Architectural Block Specifications

The hardware system is categorized into three major functional layers: memory storage structures, execution hardware, and pipeline isolation latches.

- **Register File (regbank):** Organized as 16 registers of 16-bit width, featuring dual asynchronous/synchronous read ports and a single clocked write port.
- **Data Memory (mem):** A synchronous memory space providing 256 addressable locations of 16-bit wide data words.
- **Inter-Stage Pipeline Registers:** Named sequentially as *L12*, *L23*, and *L34*, these registers are strictly bounded to alternating clock edges to enforce hazard separation.

3. Comprehensive Pipeline Stage Analysis

The complete lifetime of an instruction is distributed linearly over four processing cycles. Below is the behavioral operational map of each pipeline stage:

Stage 1: Operand Fetch & Latch (S1)

Triggered on the rising edge of the first clock phase (posedge *clk1*). The address lines *rs1* and *rs2* act as parallel indices into the register file array. The values are retrieved and stored into intermediate registers *L12_A* and *L12_B*. Concurrently, down-stream steering logic parameters (*rd*, *func*, *addr*) are passed forward into the first-stage isolation barrier.

Stage 2: ALU Execution & Operation (S2)

Triggered on the falling edge of the second clock phase (negedge clk2). The execution core hosts a highly optimized 16-bit ALU block which samples the control function L12_func to execute mathematical, relational, or shifting configurations. The resulting output vector is latched dynamically into the L23_Z holding cell.

Stage 3: Synchronous Register Write-Back (S3)

Triggered back on the rising edge of the first clock phase (posedge clk1). The pipeline takes the completed ALU calculation from L23_Z and routes it back into the register bank cell mapped by destination target index L23_rd. This ensures results are visible to incoming instruction fetches. The data is also latched into L34_Z.

Stage 4: Data Memory Commitment (S4)

Triggered on the rising edge of the second clock phase (posedge clk2). The memory write operation finishes by committing the 16-bit word held in L34_Z directly to the RAM block at index position L34_addr.

4. ALU Opcode Mapping

Opcode (func)	Operation	Hardware Expression	Functional Type
4'b0000 (0)	Addition	$L12_A + L12_B$	Arithmetic
4'b0001 (1)	Subtraction	$L12_A - L12_B$	Arithmetic
4'b0010 (2)	Multiplication	$L12_A \times L12_B$	Arithmetic
4'b0101 (5)	Bitwise AND	$L12_A \& L12_B$	Logical
4'b0110 (6)	Bitwise OR	$L12_A L12_B$	Logical
4'b1010 (10)	Logical Right Shift	$L12_A \gg 1$	Shift Operation
4'b1011 (11)	Logical Left Shift	$L12_A \ll 1$	Shift Operation

Structural Timing & Synthesis Note

All conditional assignments and register transfers utilize a non-blocking inter-assign delay parameter (#2 time-units). This ensures accurate modeling of inertial wire propagation delays and completely prevents race conditions during functional gate-level simulations.

5. I/O Port Interface Map

Signal Identifier	Bit Width	Direction	Functional Description
clk1, clk2	1 bit	Input	Two-Phase non-overlapping primary clock references.
rs1, rs2	4 bits	Input	Address line indexes for selecting source registers from the bank.
rd	4 bits	Input	Destination register index used for Stage 3 write-back operations.
func	4 bits	Input	Operational code configuration selection line for the ALU.
addr	8 bits	Input	Memory address pointer mapping directly to the 256-word SRAM array.
Zout	16 bits	Output	Continuous pipeline monitoring output tracking the L34 execution boundary.